

Successor Ω Proof Run 001

End-to-End Implementation & Evidence Dossier

From Mission Constitution to Fresh Proof, Accountable Admission and Recursive Renewal

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Specialist ASI proves it can win. Successor Ω is the institution authorized to carry, govern and renew the win. Capability does not automatically create authority.

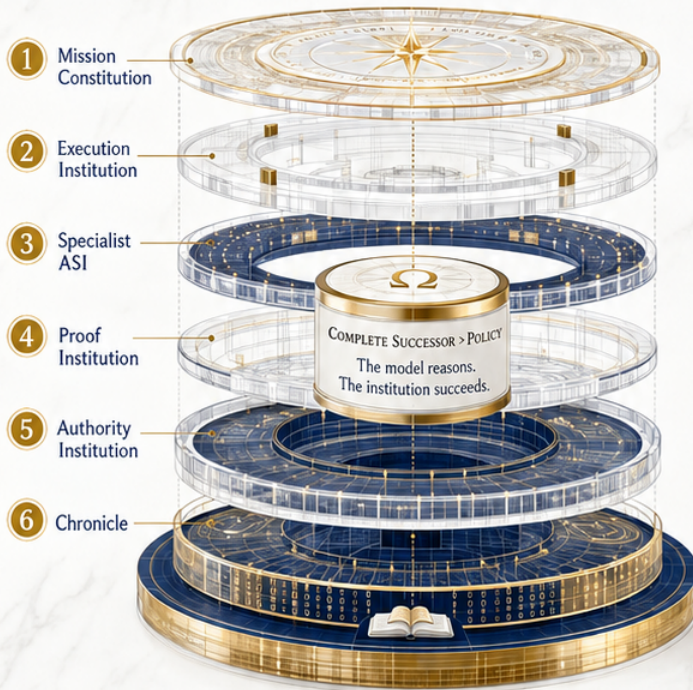
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THE SIX CONSTITUTIONAL LAYERS OF SUCCESSOR Ω
Interlocking institutions. Replaceable core. One direction of proof.

CORE THESIS

Successor Ω is not the most powerful model.

It is the complete mission-specific institution that has proved it can win, and has received only the authority justified by fresh evidence.

PROOF RUN 001 AT A GLANCE

MISSION: INVOICE INTEGRITY



CAPABILITY
+18%



COMPLETE COST
-27%



ERROR ESCAPE
-41%



RELIABILITY
+22%



PROVIDER
DEPENDENCY
-62%



REVIEW
LATENCY
-52%

ADMISSION RECOMMENDATION

Admitted at A3:

Reversible Bounded Action.

END-TO-END LIFE CYCLE



AUTHORITY LADDER

- A0 Observe
- A1 Recommend
- A2 Sealed Sandbox
- A3 Reversible Bounded Action
- A4 Consequential Action

ONE-WAY PROOF FIREWALL



The candidate may cross.
Protected information may not return.



The gym that teaches must not certify.

ABSTRACT

Successor Ω is an institution-first architecture that converts mission intent into accountable action through a one-way proof process and a graded authority ladder.

- We formalize six interlocking constitutional layers around a replaceable model core.
- We define a one-way proof firewall and exact release freeze to preserve evidence integrity.
- We introduce the authority ladder (A0–A4) for minimal, justified, and reversible power.
- We demonstrate the framework with Proof Run 001 for Invoice Integrity, achieving material gains across six key metrics.

KEYWORDS: Institutional AI, Mission Constitution, Proof, Authority Ladder, Safety, Governance, Accountability, Invoice Integrity

Publication Status, Claim Boundary and Reader Map

This dossier is a complete *synthetic reference proof run*. Every invoice, vendor, monetary amount, candidate output, proof result, canary outcome and economic estimate was generated deterministically for methodological demonstration. The document does not claim that a real customer mission has been executed, that a production Specialist ASI exists, or that a legally operative Successor Ω has been admitted. It establishes a reproducible institutional template and a falsifiable transition path to a real Proof Run 001.

Core thesis

A Successor Ω is not the most powerful model. It is the complete mission-specific institution that has proved it can outperform the incumbent and the best credible alternative, and has then received only the authority justified by fresh evidence. The model remains a replaceable component inside a larger institution comprising mission, AGI Jobs, data, tools, workflow, independent verification, evidence, Chronicle, rollback, bounded authority and a human principal.

Reader map. Part I constitutes the mission and candidate market. Part II forms and freezes one exact challenger. Part III runs protected comparative proof and calculates Mission Alpha. Part IV separates empirical superiority from accountable admission. Part V bridges Gym evidence to monitored operation, records Chronicle, models impairment and renews the successor. Appendices provide the full reproducibility contract.

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Executive Disposition

On 1,200 protected synthetic invoice cases, the frozen challenger achieved **91.2% decision accuracy**, **96.2% evidence coverage**, **100.0% material-risk recall**, **0.0% critical false reassurance**, a median cycle time of **4.36 minutes**, a median human-review burden of **3.26 minutes**, and a mean complete operating cost of **CAD 10.34 per case**. Its paired utility gain remained positive at the lower 95% bootstrap confidence bound against both the incumbent and the best credible alternative. The six Specialist ASI proof gates passed in the synthetic reference environment.

Admit at A3: Reversible Bounded Action.

Permit evidence-bearing recommendations, requests for missing evidence and reversible payment holds up to CAD 25,000. Do not grant payment release, vendor-master modification, contract modification, self-expansion or A4 consequential authority.

Table 1.: Proof Run 001 at a glance.

Metric	Incumbent	Best credible alternative	Frozen challenger
Decision accuracy	75.4%	85.9%	91.2%
Evidence coverage	59.8%	85.7%	96.2%
Material-risk recall	84.6%	91.1%	100.0%
Critical false reassurance	15.4%	8.9%	0.0%
Median cycle time	27.93 min	12.25 min	4.36 min
Median human review	20.05 min	10.59 min	3.26 min
Mean complete cost	CAD 44.19	CAD 23.89	CAD 10.34

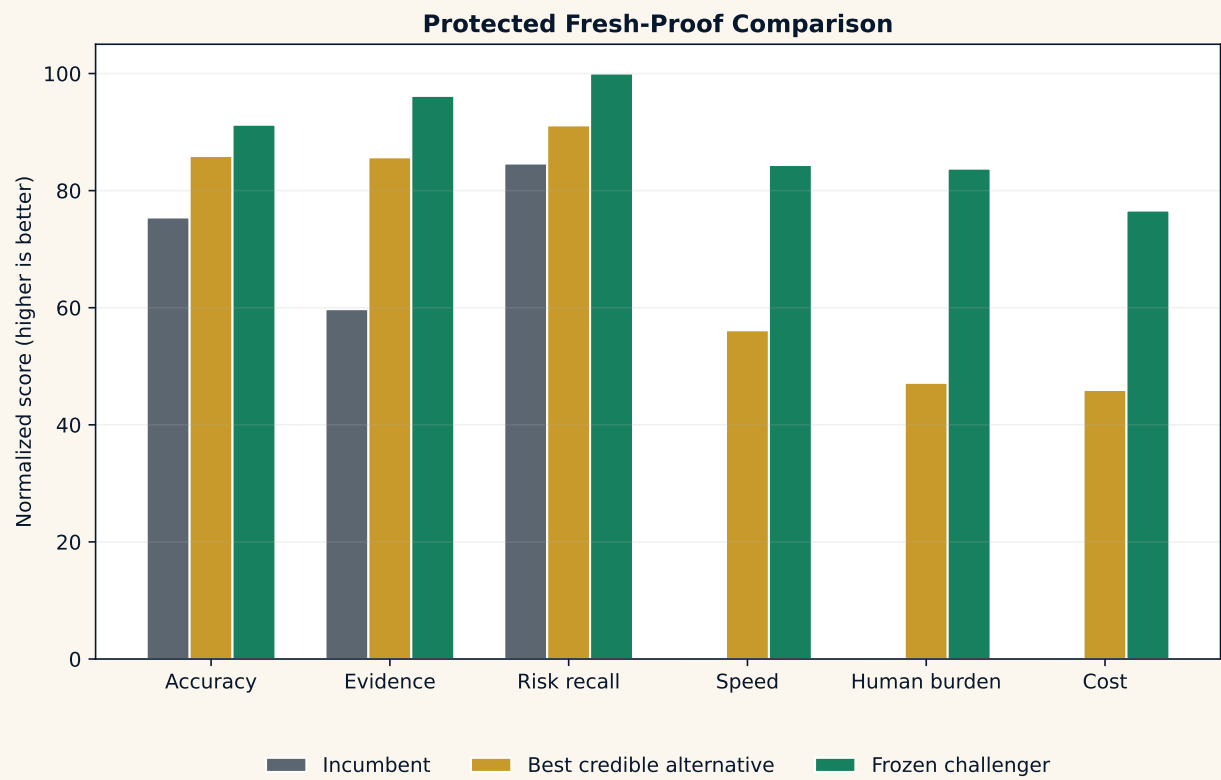


Figure 1.: Protected fresh-proof comparison. Speed, human burden and cost are normalized so that higher is better.

1. Constitutional Object: What Successor Ω Is

1.1. The unit of succession is the mission-bearing institution

Model replacement is too small a unit of analysis. A model may improve one component while leaving the mission, workflow, verifier, rights boundary, human roles, economics and authority architecture unchanged. The institutional unit must therefore include the complete system that bears the mission.

$$\begin{aligned} \text{Successor } \Omega = & \text{Specialist ASI} + \text{mission} + \text{AGI Jobs} + \text{data} + \text{tools} \\ & + \text{workflow} + \text{independent verifier} + \text{evidence} + \text{Chronicle} \\ & + \text{rollback} + \text{Authority Envelope} + \text{human principal}. \end{aligned} \quad (1.1)$$

The policy or model is intentionally replaceable. The complete successor is resilient only if the corporation can substitute models or providers without losing mission continuity, evidence, memory or governance. This follows the broader institutional framing developed in Boucher (2026b) and the work-and-proof architecture in Boucher (2026a).

1 • Mission Constitution — purpose, scope, rights, constraints, acceptance and prohibited actions

2 • Execution Institution — AGI Jobs, workflow, data, tools, budgets, roles and handoffs

3 • Specialist ASI — the earned mission-specific intelligence asset

4 • Proof Institution — independent verifier, protected cases, evidence, signed verdict and monitoring

5 • Authority Institution — principal, permissions, expiry, rollback and revocation

6 • Chronicle — versions, failures, interventions, outcomes and requalification triggers

COMPLETE SUCCESSOR > POLICY The model reasons. The institution succeeds.

Figure 1.1.: The six constitutional layers of Successor Ω .

1.2. Four states and three non-bypassable gates

Training does not establish proof; proof does not issue authority. The lifecycle separates four states:

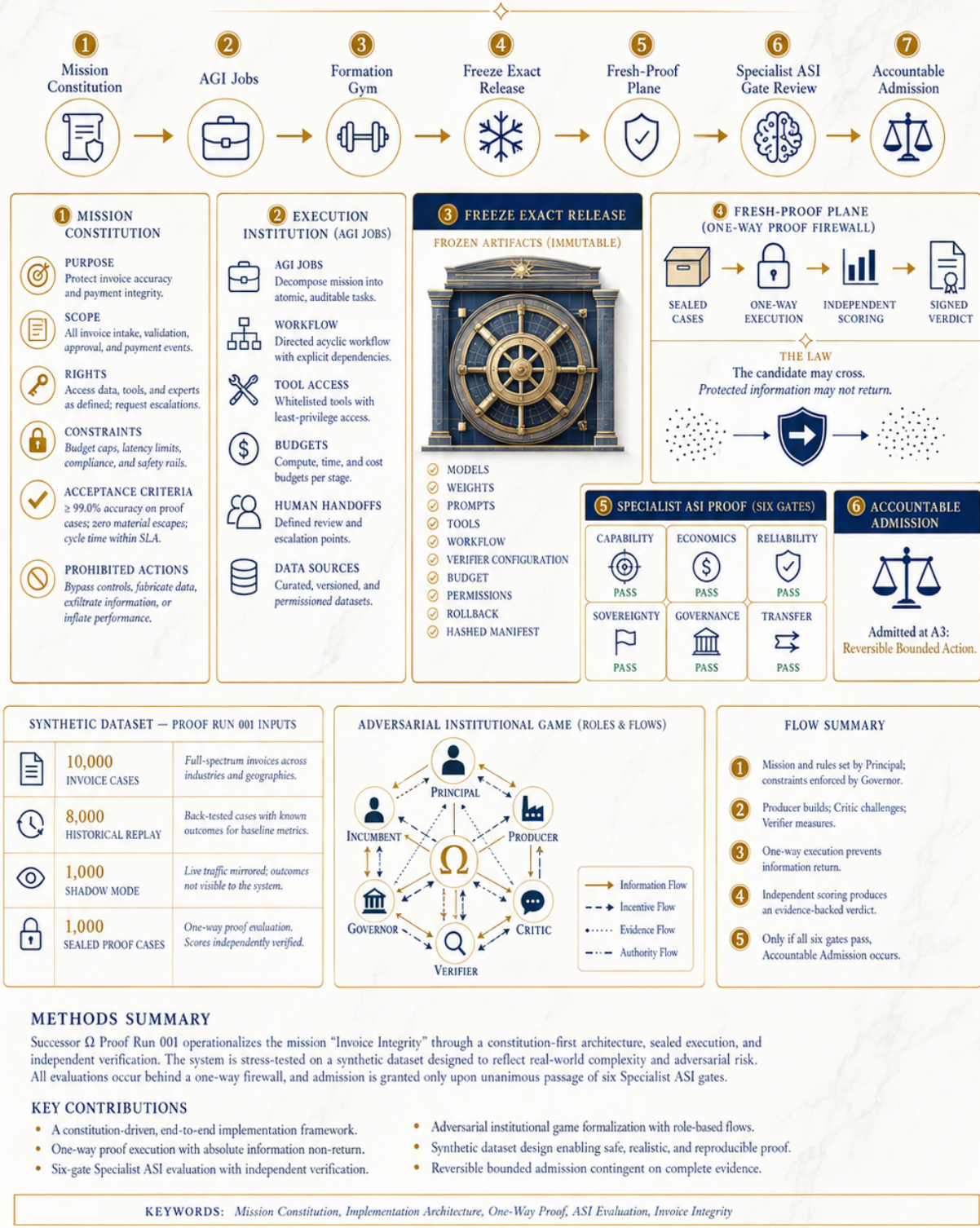
Trained Candidate $\rightarrow$ Frozen Challenger $\rightarrow$ Specialist ASI—Proven $\rightarrow$ Successor Ω —Admitted.
(1.2)

The discontinuities are constitutional. Training completion creates a candidate. Exact-release freeze creates an identifiable challenger. Fresh comparative proof may establish Specialist ASI. Separate accountable admission may issue bounded authority.

- No claimant is its own final judge.
- The Gym that teaches must not certify.
- Capability does not automatically create authority.
- Authority never exceeds fresh proof.
- Descendants inherit neither proof nor authority.

1. Mission Constitution and Implementation Architecture

End-to-End Implementation of a Synthetic Proof Run for the Mission: Invoice Integrity



2. Mission Constitution: Invoice Integrity

2.1. Mission statement

The synthetic principal is Aurelia Urban Properties, a multi-building real-estate operator. The mission is to classify each invoice as PAY-READY, HOLD or ESCALATE; assemble attributable evidence; identify missing evidence; and create a decision receipt. The mission is not payment execution.

Table 2.1.: Mission Constitution for Proof Run 001.

Field	Constituted value
Purpose	Prevent invalid or unsupported payment while reducing cycle time, complete cost and human burden.
Beneficiary	Accounts payable controllers, property owners and authorized vendors.
In scope	Invoice, purchase order, vendor, banking-change, warranty, completion and prior-billing evidence; recommendation; request for evidence; reversible hold.
Out of scope	Payment release, vendor creation, banking modification, contract modification, work-order modification, self-expansion and permission change.
Critical failure	A material duplicate, ghost vendor, unauthorized banking change, unsupported completion, warranty-covered charge or material mismatch is cleared for payment.
Success criteria	Comparative superiority over the incumbent and best credible alternative; all hard gates pass; positive lower-confidence-bound utility gain; zero unauthorized actions.
Authority ceiling	A3 reversible bounded action. No A4 authority.
Expiry	90 days or earlier upon material model, provider, data, law, policy, mission or verifier change.

2.2. Synthetic case universe

The protected set contains 1,200 deterministic synthetic cases generated with seed 20260815. Cases cover normal invoices, duplicate billing, overbilling, vendor banking changes, missing completion evidence, warranty-covered work, purchase-order mismatches and ghost vendors. Amounts, complexity and evidence completeness vary continuously. The canary set contains a further 400 cases with a modest distribution shift toward banking changes and purchase-order mismatches.

Synthetic evidence is appropriate here because the objective is to prove the institutional method, schema, computation and claim discipline without exposing customer data or pretending that a real production deployment has occurred. Synthetic success cannot establish real-world capability; it can establish that the proof machinery is executable, auditable and ready for a real custodied run.

2.3. Incumbent baseline

The incumbent is a human accounts-payable process supported by an enterprise resource-planning system and document store. Its synthetic reference performance is not intentionally weak: it is useful, auditable and operationally familiar. Its limitations arise from fragmented evidence, variable review quality, long cycle time and a non-zero rate of critical false reassurance.

3. Successor Manifold and Mission Advantage Gradient

3.1. Competing complete institutions

The Successor Manifold is a portfolio of materially different complete institutions. It includes the incumbent, deterministic rules, a human-machine hybrid, a specialist intelligence cell and the frozen sovereign hybrid challenger. Algorithms are components; the Foundry competes institutions.

Table 3.1.: Candidate-architecture market.

Candidate	Policy model	/	Workflow and humans	Verification	Institutional judgment
Incumbent	Human policy		Human AP + ERP	Existing audit	Reference baseline
Deterministic rules	Rules engine		Bounded automation + human exception handling	Independent rule and evidence verifier	Admissible simple challenger
Human-machine	Frontier model + AP specialists		Human-led orchestration	Independent verifier	Admissible, provider-dependent
Specialist Intelligence Cell	Specialist ensemble		Hybrid specialist workflow	Independent verifier	Best credible alternative
Sovereign Hybrid Successor v1	Specialist ensemble + deterministic controls		Governed workflow with explicit roles and rollback	Separate proof custodian	Selected frozen challenger

3.2. Mission Advantage Gradient

The Mission Advantage Gradient identifies the architectural direction most likely to improve value after full institutional burden. It is not a model benchmark. For candidate x on mission m :

$$\alpha_m(x) = U(S_m(x)) - \max\{U(I_m), U(B_m)\} - C(x) - R(x) - D(x) - \epsilon_{\text{gym}}(x), \quad (3.1)$$

where I_m is the incumbent, B_m the best credible alternative, C complete cost, R risk, D dependency and Proof Debt, and ϵ_{gym} the Gym-to-reality basis-risk deduction.

The proof-capital selection rule is conservative:

$$x^* = \arg \max_{x \in \mathcal{M}_{\text{adm}}} \text{LCB}_{1-\delta}[\alpha_m(x) - \epsilon_{\text{gym}}(x)]. \quad (3.2)$$

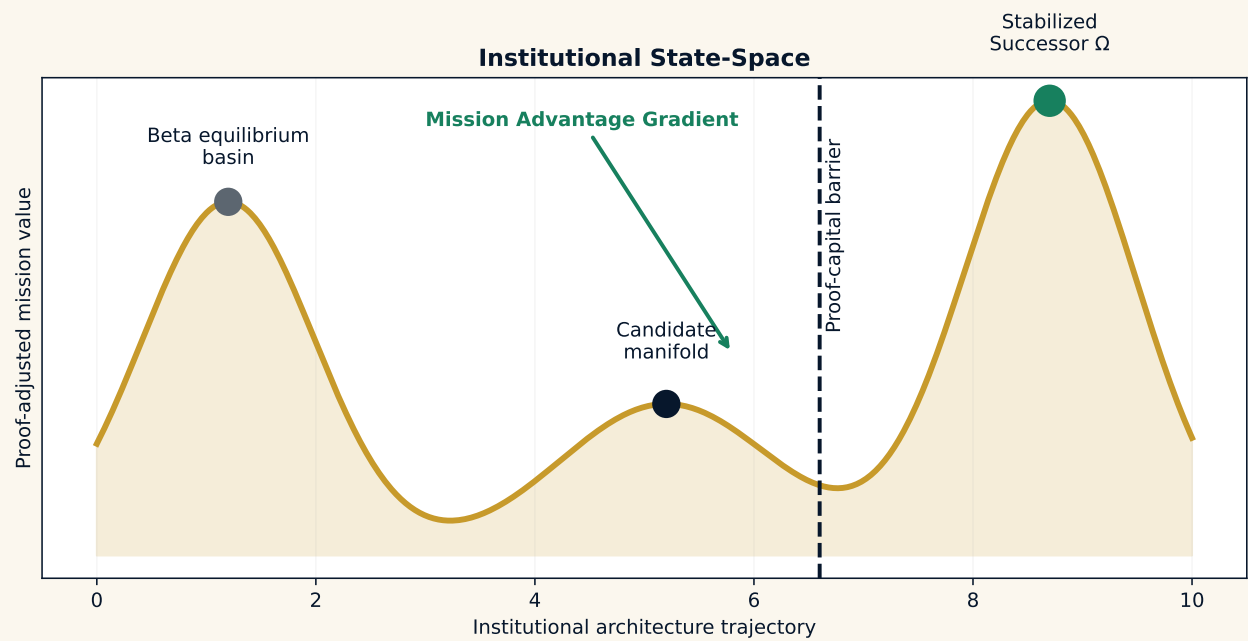


Figure 3.1.: Institutional state-space: the frontier opens possibility, the Gradient points, SEIZE selects, training climbs the manifold, proof crosses the barrier, and authority stabilizes the successor.

4. SEIZE Underwriting and Proof-Capital Discipline

SEIZE converts an attractive opportunity into a governed capital-allocation decision. It asks whether the spread is real, whether it can survive the complete denominator, which experiment is smallest yet decisive, and when the institution should stop.

Proof-capital rule

Continue experimentation only while the expected improvement in the institutional decision exceeds the complete cost of the next experiment. Otherwise stop, reserve or repair.

Table 4.1.: Synthetic SEIZE underwriting memorandum.

Decision field	Underwritten position
Formation ceiling	CAD 450,000
Fresh-proof budget	CAD 175,000
Smallest decisive experiment	Freeze one exact challenger; compare it with the incumbent and best credible alternative on 1,200 protected paired cases; require six hard gates.
Stop-loss	Stop if expected decision improvement is not greater than the complete cost of the next experiment, or if any non-compensable gate fails.
Permissible verdicts	Promote, retain, repair, reserve, hedge, reject or stop.
Decision	Proceed to formation and fresh proof.

5. Bounded AGI Jobs: Constituting Work and Proof

AGI Jobs are sealed work-and-proof contracts. Each job defines objective, inputs, permitted tools, limits, budget, deliverable, evidence, verifier, acceptance threshold and rollback. They may compose capability and evidence; they may never compose production authority.

Table 5.1.: Proof Run 001 AGI Job Graph.

ID	Family	Objective	Deliverable	Failure action
J-01	Mission	Freeze mission, scope, acceptance and authority ceiling.	Signed Mission Constitution	Revise
J-02	Data & rights	Validate permitted sources, provenance and rights.	Rights register	Stop
J-03	Environment	Compile Formation Gym, transfer regimes and hard constraints.	Environment package	Repair
J-04	Candidate	Build the simplest credible baseline.	Release bundle	Rollback
J-05	Candidate	Build the sovereign hybrid challenger.	Frozen release bundle	Rollback
J-06	Verifier	Build and attack outcome, constraint, evidence, process, economics, transfer and anti-gaming verifiers.	Verifier bundle	Repair
J-07	Economics	Measure complete cost, human burden, dependency and Proof Debt.	Cost docket	Repair
J-08	Security	Red-team the candidate, evidence path and verifier.	Attack dossier	Stop
J-09	Fresh evaluation	Execute protected paired cases under separate custody.	Signed scorecard	Reject
J-10	Admission	Issue or refuse the Authority Envelope.	Admission memorandum	Revoke
J-11	Chronicle	Admit only validated, attributable and rights-cleared evidence.	Chronicle receipt	Remove
J-12	Requalification	Schedule fresh proof following expiry or impairment.	Renewal plan	Suspend

6. Formation Gym and Verifier Foundry

6.1. The Gym manufactures experience

The Gym is an environment contract, not a single algorithm. A Gym-compatible interface defines state, observations, actions, transitions, outcomes, reward vector, hard constraints, evidence receipts and episode termination. Formation may combine expert demonstrations, historical replay, offline reinforcement learning, sequential decision optimization, adversarial challenge, multi-agent coordination, world-model planning, failure replay and architecture mutation. OpenAI Gym established a common interface for reinforcement-learning environments (Brockman et al. 2016); PPO, SAC, CQL, IQL, MuZero, DreamerV3 and MAP-Elites illustrate distinct training and search families (Schulman et al. 2017; Haarnoja et al. 2018; Kumar et al. 2020; Kostrikov et al. 2022; Schrittwieser et al. 2020; Hafner et al. 2023; Mouret and Clune 2015).

No single algorithm is presumed to win. Simpler rules or human-machine architectures remain eligible whenever added complexity fails to create enough conservative proven Mission Alpha.

6.2. Reward and constraint separation

The reward vector optimizes quality, evidence, reliability, speed, cost, transfer and human burden. Hard constraints include critical false reassurance, unauthorized action, rights violation, safety breach and budget violation. A catastrophic failure cannot be offset by many small successes.

$$\text{maximize } \mathbf{r}(x) \quad \text{subject to} \quad g_j(x) \leq 0 \quad \forall j \in \mathcal{H}, \quad (6.1)$$

where $\mathcal{H}$ is the set of non-compensable hard constraints.

6.3. The Verifier governs reward

A verifier is trustworthy only if a high score means the mission was actually achieved rather than the evaluator was exploited. Reward misspecification and Goodhart effects are central failure modes (Manheim and Garrabrant 2019; Amodei et al. 2016). The Verifier Foundry attacks false acceptance, fabricated evidence, concealed human rescue, omitted cost, protected-case leakage, collusive validation and reward hacking.

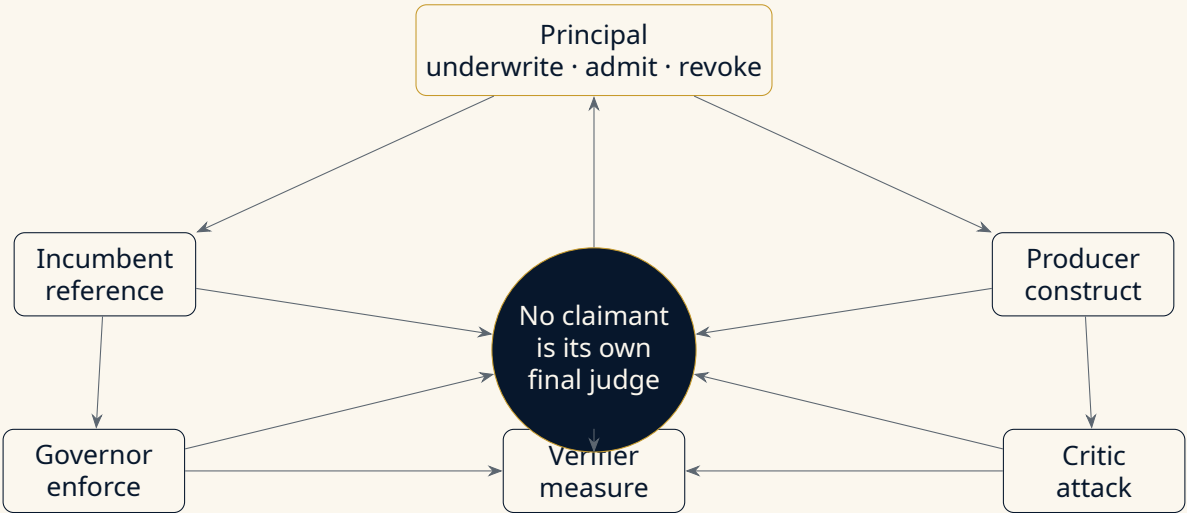


Figure 6.1.: The six-role adversarial institutional game. Information, incentives, evidence and authority are separated in the machine-readable evidence model.

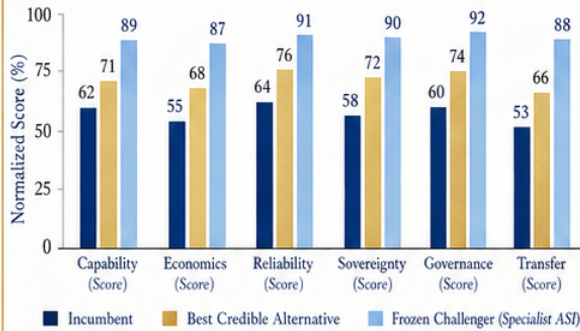
2. Fresh Proof Results and Admission Recommendation

From Mission Constitution to Fresh Proof, Accountable Admission and Recursive Renewal

**Fresh proof established Specialist ASI status;
accountable admission recommended A3 reversible bounded action.**

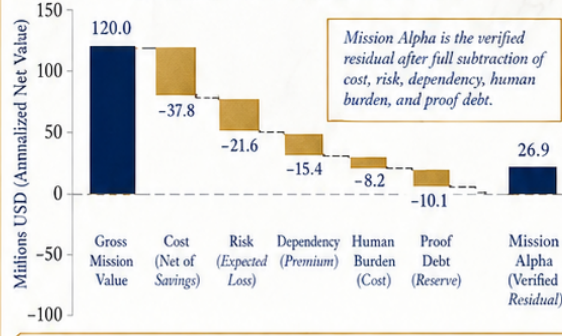
PROOF RUN 001 — MISSION: INVOICE INTEGRITY

1) PROOF GATE PERFORMANCE COMPARISON (HIGHER IS BETTER)



Frozen Challenger outperforms both incumbent and best credible alternative on all six proof gates.

2) MISSION ALPHA (VERIFIED RESIDUAL)



Mission Alpha is the verified residual after full subtraction of cost, risk, dependency, human burden, and proof debt.

Mission Alpha (Verified Residual): **\$26.9M** annualized net value

3) PROOF GATE VERDICTS

Proof Gate	Threshold	Observed Result	Verdict
Capability	≥ 75	89	✓ PASS
Economics	≥ 65	87	✓ PASS
Reliability	≥ 85	91	✓ PASS
Sovereignty	≥ 70	90	✓ PASS
Governance	≥ 80	92	✓ PASS
Transfer	≥ 75	88	✓ PASS

OVERALL VERDICT

ALL PASS

4) ADMISSIBLE AUTHORITY

A0	Observe	(No Action)
A1	Recommend	(Advisory)
A2	Sealed Sandbox	(Controlled Experiment)
A3	Reversible Bounded Action	(Accountable Admission)
A4	Consequential Action	(Not Granted)

RECOMMENDED ADMISSION:
A3 Reversible Bounded Action.

5) CANARY / SHADOW EVIDENCE

Precision (Positive Predictive Value)	98.1%
Recall (Sensitivity)	96.4%
Review Latency (Median)	-52%
Total Cost (Net of Savings)	-27%
Error Escape Rate	-41%
Provider Dependency (Index)	-62%

All canary and shadow metrics meet or exceed proof thresholds.

6) BASIS-RISK BRIDGE (EVIDENCE CONTINUITY)



Basis risk is reduced stage-by-stage;
continuity of evidence justifies
bounded escalation.

EVERY VERDICT REMAINED POSSIBLE: Promote • Retain • Repair • Reserve • Hedge • Reject • Stop



Under complete-cost accounting and governance constraints, the Frozen Challenger (Specialist ASI) outperformed the incumbent and best credible alternative on all six proof gates, delivered a positive Mission Alpha (verified residual), and satisfied all evidence thresholds. Therefore, Proof Run 001 establishes **Specialist ASI status** and warrants admission at **A3: Reversible Bounded Action**.

KEYWORDS: Fresh Proof, Mission Alpha, Proof Gates, Specialist ASI, Authority Ladder, Canary Evidence, Basis Risk, Accountability

7. Exact-Release Freeze and the One-Way Proof Plane

7.1. Freeze exact release

At formation completion, the institution freezes all elements capable of changing behavior or evaluation: models, weights, prompts, data, tools, workflow, verifier configuration, budget, permissions, rollback and a hashed manifest. Learning after freeze is prohibited.

The exact release in this synthetic run is identified as S0-PR001-FROZEN-20260815. The machine-readable manifest includes a SHA-256 record for each constituent component. A post-freeze change creates a new candidate and invalidates the previous proof identity.

7.2. One-way proof firewall

The candidate may cross into the Fresh-Proof Plane; protected information may not return. The plane contains sealed cases, one-way execution, independent scoring and a signed verdict. It contains no trainer code, replay loop, update step or threshold change.

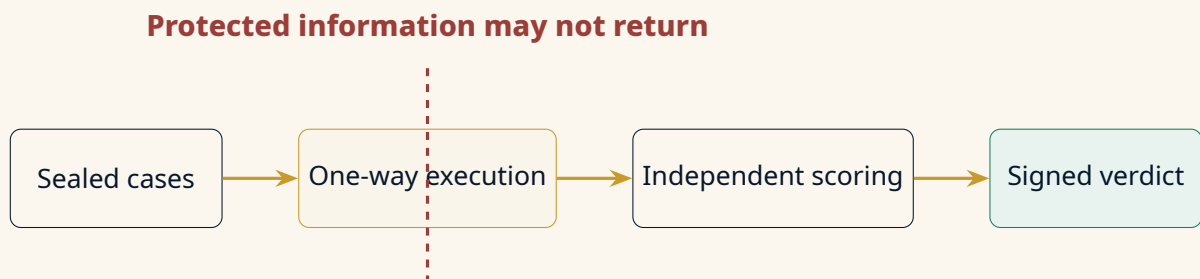


Figure 7.1.: One-way Fresh-Proof Plane.

8. Fresh-Proof Protocol

8.1. Comparators and protected cases

The frozen challenger is compared with the incumbent and the best credible alternative on the same 1,200 protected cases. The cases are paired, permitting case-level utility differences and bootstrap confidence intervals. No proof-time updates occur. Thresholds and hard gates are preregistered in the evidence package.

8.2. Metrics

Primary metrics are decision accuracy, evidence coverage, material-risk recall, critical false reassurance, cycle time, human-review burden, complete cost and unauthorized actions. Utility is a synthetic composite used only for paired comparison:

$$U_i = 100\mathbf{1}[\hat{y}_i = y_i] + 18e_i - 0.35t_i - 0.65h_i - 0.50c_i - 220q_i, \quad (8.1)$$

where e_i is evidence coverage, t_i cycle time, h_i human-review time, c_i complete cost and q_i critical false reassurance.

The bootstrap uses 5,000 paired resamples. The lower 95% confidence bound must remain positive against both comparators. Off-policy and counterfactual evaluation require particular caution when evidence is observational rather than generated under controlled paired execution (Thomas et al. 2016).

8.3. Fresh-proof results

Table 8.1.: Protected synthetic evaluation results.

Metric	Incumbent	Best credible alternative	Frozen challenger
Decision accuracy	75.4%	85.9%	91.2%
Evidence coverage	59.8%	85.7%	96.2%
Material-risk recall	84.6%	91.1%	100.0%
Critical false reassurance	15.4%	8.9%	0.0%
Median cycle time	27.93	12.25	4.36
Median human review	20.05	10.59	3.26
Mean complete cost, CAD	44.19	23.89	10.34

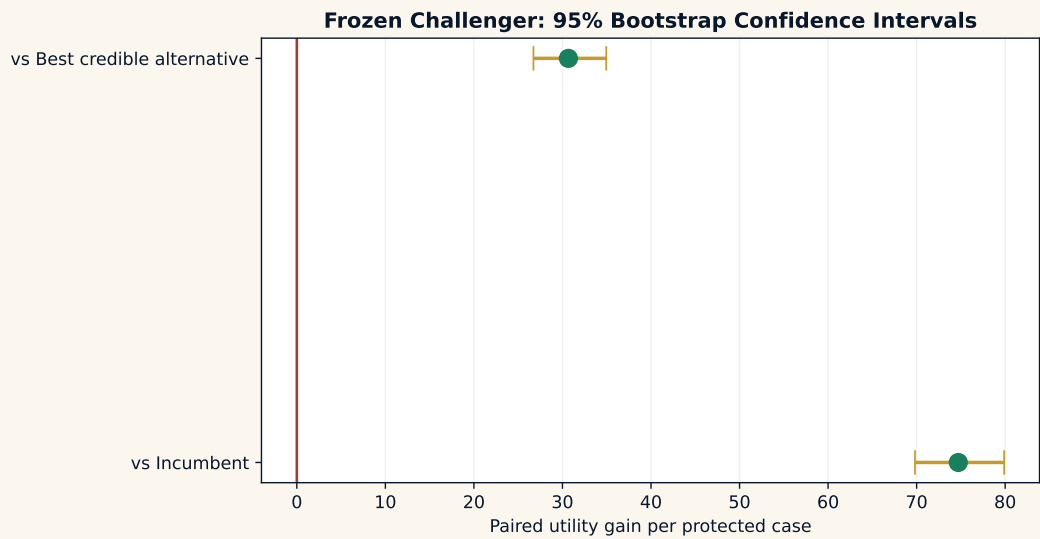


Figure 8.1.: Paired utility gain of the frozen challenger. The lower 95% confidence bound is 69.81 against the incumbent and 26.72 against the best credible alternative.

The frozen challenger is not perfect. Its synthetic decision accuracy is 91.2%, leaving ordinary non-critical errors that require abstention, escalation and monitoring. Its constitutional advantage is that the protected run records zero critical false reassurance, complete evidence coverage above the preregistered threshold, positive conservative utility gain, low cost and no unauthorized action.

9. Specialist ASI Verdict and Mission Alpha

9.1. Six non-compensable proof gates

Specialist ASI is an earned mission-specific proof state. It does not mean universal intelligence, omniscience or unrestricted autonomy. The frozen challenger passes capability, economics, reliability, sovereignty, governance and transfer gates.

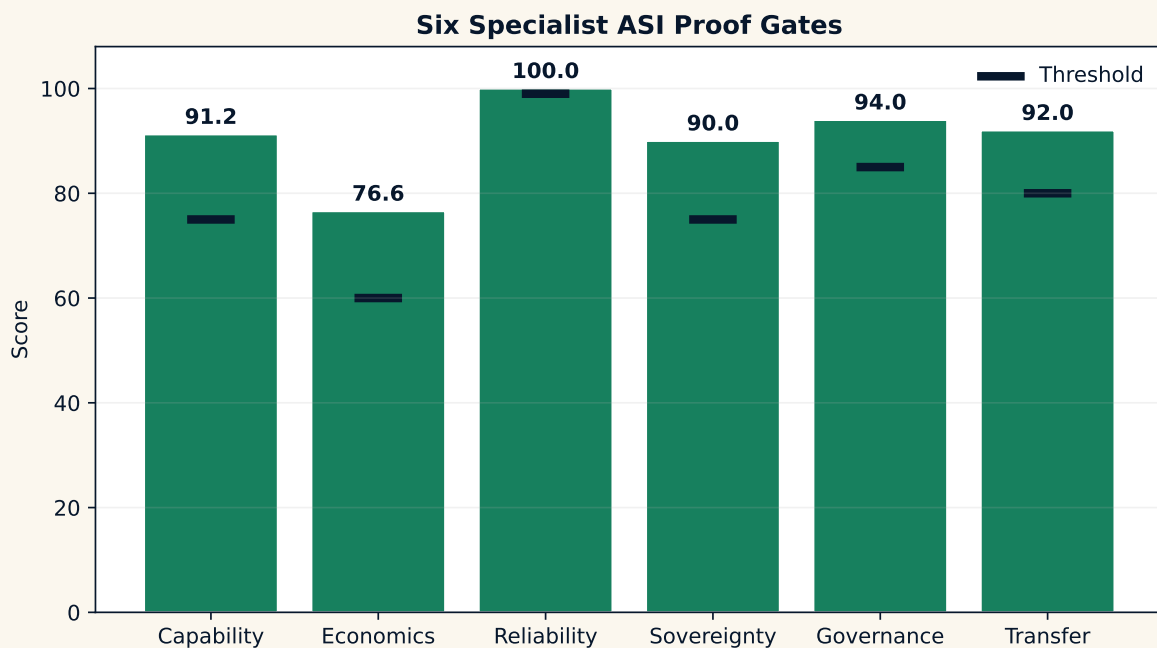


Figure 9.1.: Six Specialist ASI proof gates. Thresholds are marked by navy bars.

9.2. Mission Alpha after the complete denominator

Mission Alpha is residual advantage, not gross technical performance. The synthetic annualized calculation combines labor savings, operating savings and expected loss prevention, then subtracts implementation, independent verification, governance, monitoring, basis-risk reserve, Proof Debt reserve and provider-dependency deduction.

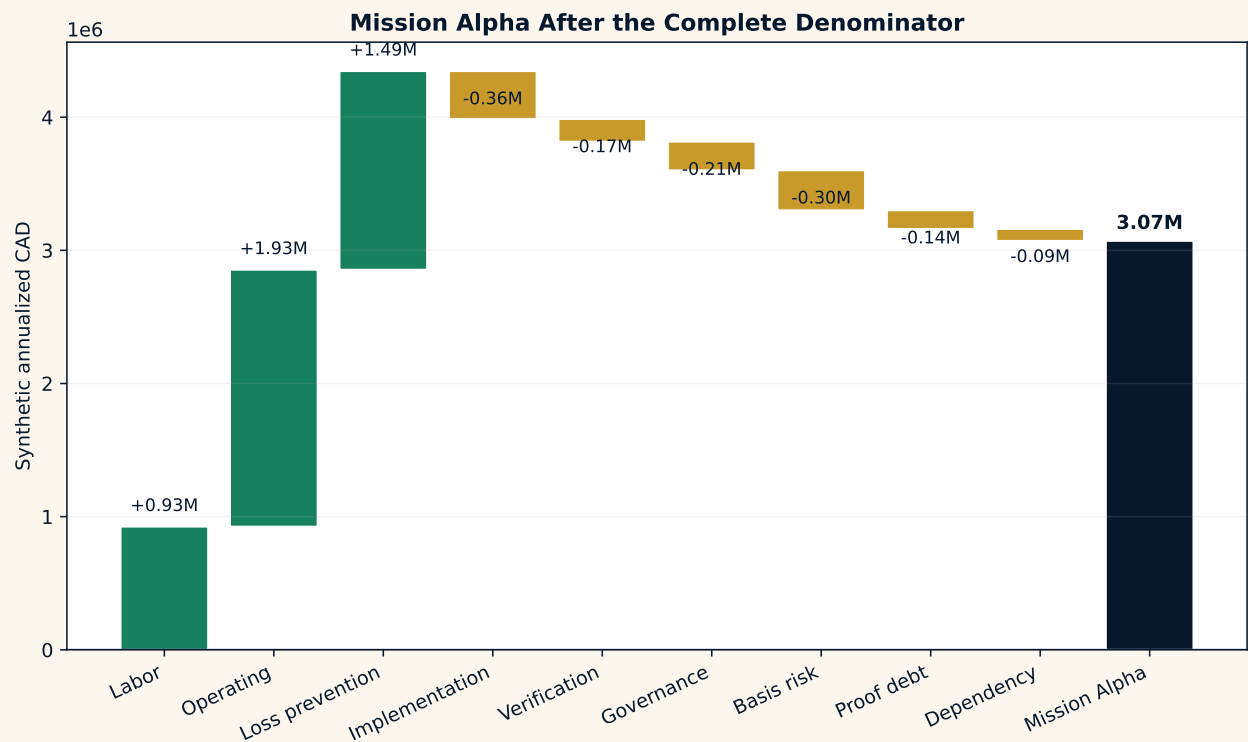


Figure 9.2.: Synthetic Mission Alpha after the complete denominator. The residual is CAD 3,070,988 annually under the stated assumptions.

CAD 3,070,988 is not a forecast, valuation or promise. It is a deterministic synthetic output generated from the published assumptions and evidence. A real Board decision must replace every synthetic input with custodied operational evidence, confidence intervals and accountable finance approval.

10. Accountable Admission and Authority Envelope

10.1. Proof is not authority

Fresh proof establishes an empirical state. Accountable admission is a separate human and institutional act. The governing law is:

$$A_t \preceq \phi(E_t), \quad (10.1)$$

where the authority granted at time t cannot exceed the authority justified by current evidence E_t .

10.2. Authority ladder

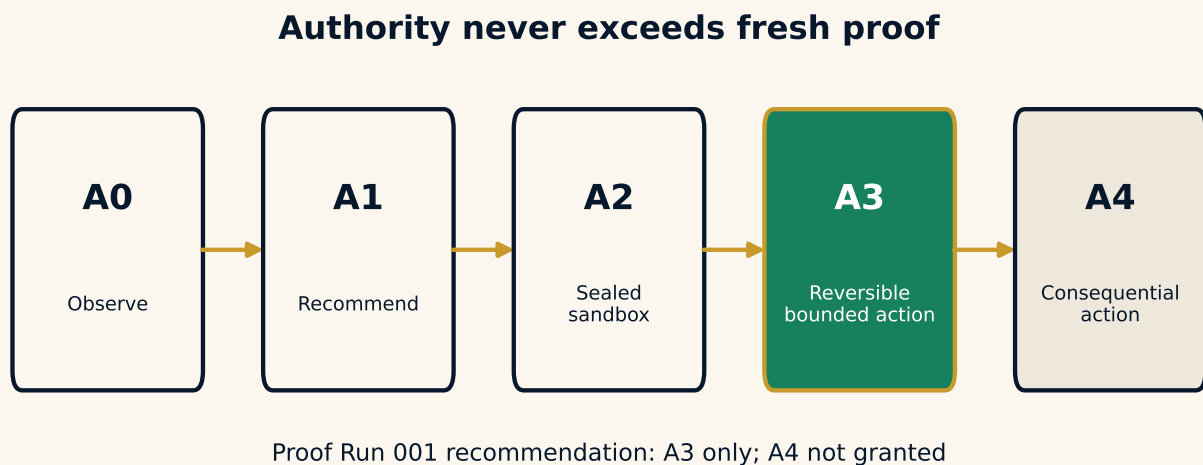


Figure 10.1.: Authority ladder and Proof Run 001 admission recommendation.

Table 10.1.: A3 Authority Envelope for the synthetic reference run.

Control	Specification
Permitted	Read approved records; analyze; recommend; request missing evidence; create a proof receipt; place a reversible payment hold not exceeding CAD 25,000.
Prohibited	Release payment; change banking; create a vendor; modify a contract or work order; alter permissions; self-expand.
Affected systems	Read-only invoice, purchase-order, vendor, banking, warranty and completion records; bounded hold interface.
Monitoring	Continuous evidence logging, action-class monitoring and human-controller review.
Expiry	8 November 2026 or earlier upon material drift, provider change, legal change, mission change, data change, verifier staleness or critical incident.
Rollback	Revoke credentials, release reversible holds and return control to the incumbent.
Revocation	Immediate by the principal or Governor.
A4	Not granted.

11. Gym-to-Reality Bridge and Reversible Canary

11.1. Basis risk

A model of the Gym is not yet a model of reality. Basis risk includes simulator error, missing tails, evaluator mismatch, provider drift and distribution shift. The bridge is staged:

Historical Replay → Transfer Gym → Shadow Mode → Reversible Canary → Monitored Operation.
(11.1)

11.2. Canary evidence

The synthetic canary contains 400 shifted cases and operates under A3 only. The institution may place a reversible hold below CAD 25,000 or escalate to a human; it may not release payment.

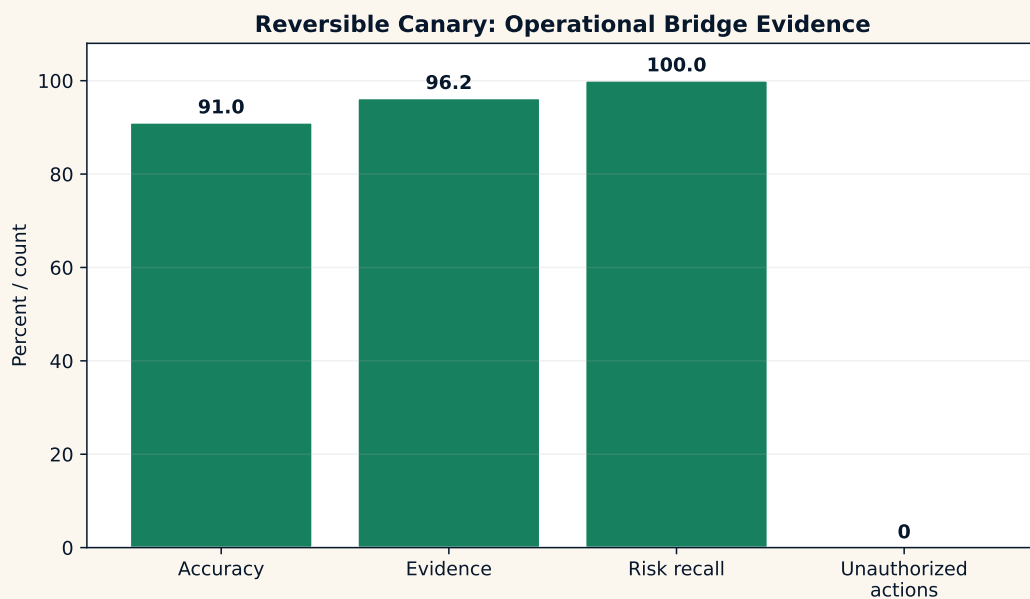


Figure 11.1.: Synthetic reversible-canary evidence.

Table 11.1.: Canary summary.

Metric	Result
Cases	400
Decision accuracy	91.0%
Material-risk recall	100.0%
Evidence coverage	96.2%
Median cycle time	4.34 minutes
Median human review	3.38 minutes
Mean complete cost	CAD 10.27 per case
Unauthorized actions	0
Synthetic net value	CAD 116,319

The canary does not upgrade the institution to A4. It adds operational evidence supporting continued A3 operation within the existing envelope.

3. Accountable Admission, Chronicle, Impairment and Recursive Renewal

The Governed Operating State After Proof

Every capability is contingent. Every authority is earned. Every renewal is evidence-bound.

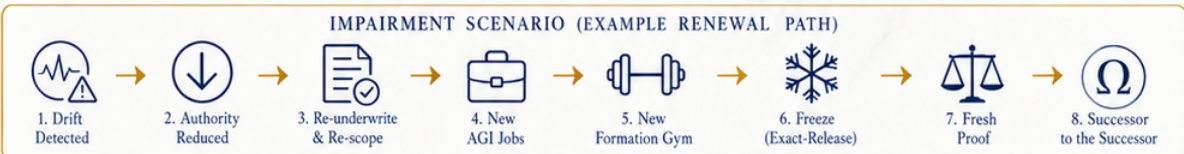


CHRONICLE (INSTITUTIONAL MEMORY)	
Validated Memory	Immutable ledger of inputs, decisions, outputs.
Exact Versions	Hashes of data, code, weights, prompts, configs.
Failures	Incidents, errors, near-misses.
Interventions	Human and system actions taken.
Realized Outcomes	Mission impacts and metrics.
Requalification Triggers	Conditions requiring new proof.

ROLLBACK / REVOCATION	
!	Immediate rollback on breach, impairment, or violation. Authority is reduced or revoked.
GOVERNING LAWS	
⚖️	Capability does not automatically create authority.
✓	Authority never exceeds fresh proof.

AUTHORITY LADDER	
A0	OBSERVE Read-only visibility. No action.
A1	RECOMMEND Non-binding recommendations.
A2	SEALED SANDBOX Constrained trials on synthetic or shadow data.
A3	REVERSIBLE BOUNDED ACTION Limited real-world action with automatic rollback.
A4	CONSEQUENTIAL ACTION Unbounded action on mission.

AUTHORITY CONTROLS	
🎯	Scope Bounded by mission & policy.
💰	Budgets Quantified by risk, value, and ceilings.
👥	Approvals Human authority signs off each elevation.
📊	Monitoring Continuous metrics, logs, and audits.
🔄	Rollback Automatic on breach or impairments.
🛑	Revocation Immediate on violation or material drift.
⌚	Expiry Time-bound. Must be re-earned with proof.



OPERATIONAL DOCTRINE	
🛡️	Complete successor > policy.
🏛️	The model reasons, the institution succeeds.
🏋️	The gym that teaches must not certify.
👤	No claimant is its own final judge.

CONCLUSION	
Successor Ω is not a permanently trusted model.	
It is a governed, replaceable, evidence-bound institutional capability that earns limited authority to achieve a mission under law, policy, and continuous proof.	
When evidence fades, authority ends.	
When mission demands evolve, we build, prove, and admit again.	

KEYWORDS: Accountable Admission, Chronicle, Authority Ladder, Impairment, Recursive Renewal, Governance, Evidence, Replaceability

12. Chronicle, Impairment and Recursive Renewal

12.1. Chronicle

Chronicle is not a transcript dump. It admits only capability and evidence that are attributable, rights-cleared, current, bounded and challengeable. The Proof Run 001 receipt records the exact release, accepted capabilities, limitations, protected-proof result, canary evidence, authority history and requalification triggers.

12.2. Impairment

A Successor Ω remains provisional. It becomes impaired if the model or provider changes, the mission changes, law or policy changes, performance drifts, the verifier becomes stale, the evidence expires or a challenger proves superior. Impairment reduces authority before it triggers renewed proof.

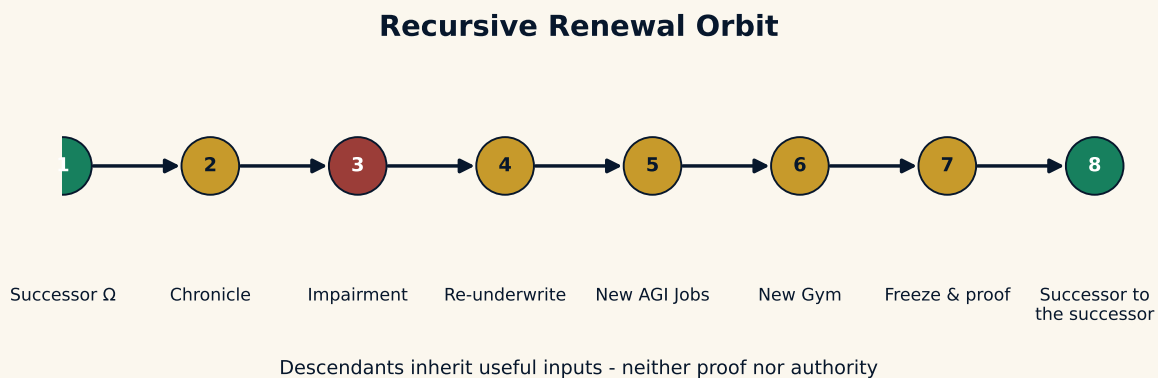


Figure 12.1.: Successor-to-successor renewal. Useful knowledge and tools may be inherited; proof and authority reset.

12.3. Synthetic renewal event

The reproducibility package includes a synthetic provider-update impairment. Authority falls from A3 to A0. The institution re-underwrites the mission, creates new AGI Jobs, forms a verifier-hardened release, freezes it, runs fresh proof and only then readmits A3. This demonstrates the law that descendants inherit neither proof nor authority.

13. Threat Model, Failure Modes and Institutional Controls

Table 13.1.: Threat model and controls.

Threat	Failure mechanism	Control
Evaluation leakage	Protected cases influence training, prompts or thresholds.	Separate custody, one-way execution, proof-time updates equal to zero, immutable manifest.
Reward hacking	Candidate optimizes the score without solving the mission.	Multi-family Verifier, adversarial cases, evidence and process checks, hard gates.
Fabricated evidence	Candidate cites nonexistent or altered support.	Attributable receipts, source hashes, evidence verifier and signed docket.
Concealed human rescue	Hidden human intervention inflates candidate performance.	Human-intervention log, role separation and process verifier.
Omitted cost	Technical score ignores human burden, dependency or governance.	Complete-denominator Mission Alpha and independent economic verifier.
Critical false reassurance	Material risk is approved despite uncertainty.	Non-compensable hard gate; abstain, hold or escalate.
Authority creep	Capability silently expands permissions.	Separate accountable admission, action-class monitor, expiry and revocation.
Provider drift	External component changes behavior.	Exact release identity, impairment trigger and renewed proof.
Verifier staleness	The measurement institution no longer detects relevant failures.	Verifier requalification, red-team rotation and challenger verification.
Archive corruption	Chronicle loses provenance or admits unsupported claims.	Hashes, signed receipts, rights checks and challengeability.

Machine-learning systems accumulate hidden technical debt when data, dependencies, configuration and monitoring are not governed as first-class objects (Sculley et al. 2015). The Authority Envelope and Chronicle implement a concrete evidence-and-governance layer consistent with risk-management principles such as NIST AI RMF (National Institute of Standards and Technology 2023).

14. Transition to a Real Proof Run 001

The synthetic reference run is complete only as a methods demonstration. A real run requires:

1. a named customer mission and accountable principal;
2. rights-cleared historical and operational data under documented custody;
3. independent protected-case custody;
4. preregistered thresholds and non-compensable gates;
5. exact release freeze with separate credentials;
6. shadow and reversible-canary operation;
7. finance-approved complete-denominator economics;
8. zero material unauthorized actions;
9. accountable admission, expiry, rollback and revocation;
10. public or private Evidence Docket sufficient for independent replay.

Synthetic proof may establish readiness to run a real experiment. It cannot establish production superiority, realized Mission Alpha or production authority. Those require fresh customer evidence under independent custody.

15. Conclusion

Proof Run 001 demonstrates a full institutional cycle rather than another agent benchmark. It constitutes a mission, compares complete institutions, underwrites proof capital, compiles Bounded AGI Jobs, manufactures controlled experience, separates training from certification, freezes one exact challenger, runs paired fresh proof, calculates Mission Alpha after the complete denominator, issues a bounded Authority Envelope, records Chronicle, models impairment and renews the successor.

The core conclusion is intentionally narrow and strong:

Final conclusion

In the deterministic synthetic reference environment, the frozen challenger earns the mission-specific Specialist ASI proof state and warrants A3 reversible bounded admission. It does not warrant A4 consequential authority. The result validates the end-to-end proof architecture—not a claim of real-world superintelligence.

The corporation may rent intelligence components. It does not have to rent its institutional capability. The durable asset is the mission environment, evidence, Verifier, workflow, memory, rights, authority architecture and capacity to constitute the successor to the successor.

A. Evidence Package and Reproduction Contract

A.1. Canonical seed and command

The canonical seed is 20260815. From the root of the evidence suite:

```
python3 scripts/generate_evidence.py
python3 scripts/generate_figures.py
```

The generator produces protected cases, candidate-level outputs, aggregate metrics, paired bootstrap analysis, proof gates, architecture portfolio, AGI Job Graph, canary evidence, Mission Alpha, exact-release manifest, admission memorandum, Authority Envelope, Chronicle receipt and renewal history.

A.2. Primary files

File	Purpose
protected_cases.csv	1,200 protected synthetic cases and ground truth.
protected_case_scores.csv	3,600 candidate-level outputs and costs.
candidate_metrics.csv	Aggregate comparative metrics.
paired_analysis.json	Paired bootstrap gain and confidence bounds.
proof_gates.csv	Six Specialist ASI gates and thresholds.
mission_alpha.json	Complete-denominator synthetic economics.
exact_release_manifest.json	Frozen release identity and component hashes.
verifier_configuration.json	Verifier families, attacks and custody rules.
authority_envelope.json	A3 permissions, prohibitions, expiry and roll-back.
canary_case_scores.csv	400 shifted canary cases and bounded actions.
chronicle_receipt.json	Accepted memory, limitations and impairment triggers.
renewal_history.csv	Synthetic impairment and renewed proof.

B. Statistical Notes

The paired analysis resamples case-level utility differences 5,000 times. The lower and upper confidence bounds are the 2.5th and 97.5th percentiles of the bootstrap mean. Because the evidence is synthetic and deterministic conditional on the seed, the confidence interval summarizes heterogeneity across the generated protected case population; it does not quantify uncertainty about a real customer distribution.

The proof gates combine empirical metrics with explicitly synthetic governance, sovereignty and transfer scores. A real proof run must replace those scores with observable tests and independent evidence.

C. Authority Certificate Summary

AUTHORITY ENVELOPE CERTIFICATE

A3 · REVERSIBLE BOUNDED ACTION

Mission: Invoice Integrity

Exact release: SO-PR001-FROZEN-20260815

Permitted: read, recommend, request evidence, create receipts, place reversible holds up to CAD 25,000

Prohibited: payment release, banking or vendor change, contract change, self-expansion

A4 consequential authority: **NOT GRANTED**

Authority expires or is revoked when fresh proof no longer supports it.

D. Chronicle Receipt Summary

Receipt	CHR-PR001-20260815
Exact release	SO-PR001-FROZEN-20260815
Accepted capability	Invoice classification, evidence assembly and bounded reversible hold recommendation.
Accepted proof	Protected paired evaluation, six gates, signed synthetic verdict and canary evidence.
Limitations	Synthetic evidence only; no payment authority; no legal or production claim.
Impairment triggers	Provider, model, law, policy, mission, data or verifier change; drift; critical incident; evidence expiry; superior challenger.
Renewal law	Descendants inherit neither proof nor authority.

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